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Universal Child Care, Maternal Labor Supply, and Family Well-Being

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We analyze the introduction of highly subsidized, universally accessible child care in Quebec, addressing the impact on child care utilization, maternal labor supply, and family well-being. We find strong evidence of a shift into new child care use, although some crowding out of existing arrangements is evident. Maternal labor supply increases significantly. Finally, the evidence suggests that children are worse off by measures ranging from aggression to motor and social skills to illness. We also uncover evidence that the new child care program led to more hostile, less consistent parenting, worse parental health, and lower-quality parental relationships.

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There has been a substantial rise in the share of mothers who work in the paid labor force in North America over the past generation. In the United States, the share of mothers with children under age 6 who work rose from 34 percent in 1976 to 56 percent in 2004. In Canada, the employment rate of mothers with at least one child younger than 6 rose from 31 percent in 1976 to 67 percent in 2004. In neither country has this trend been offset by a decline in the proportion of working fathers, necessitating an increased use of paid and unpaid child care. In 1984, 37 percent of children in the United States under age 6 were being cared for by someone other than a parent; by 2001 the proportion had increased to 56 percent. In Canada, the comparable percentages rose from 40 percent in 1994–95 to 51 percent in 2002–3.¹

The increased demand for child care accompanying the rise of two-earner couples is attracting the attention of public policy makers, with good reason: the vast majority of children in nonparental care come from two-parent families.² Currently, in both Canada and the United States, larger child care subsidies are targeted at low-income families; middle- and upper-income families receive modest tax subsidies for either child care or preschool. But interest is growing in more universal subsidies for early child care/education, as offered in many nations in Europe (OECD, various years). In Canada, the province of Quebec introduced universal subsidies for child care over the 1997–2000 period, and a central debate of the 2005–6 federal election was a plan for a national child care program. In the United States, universal preschool initiatives have been passed by states such as Georgia, New York, and Oklahoma. The controversy over California's Proposition 82 reveals the conflict this issue provokes. Unfortunately, many of these debates continue without the benefit of evidence on the consequences of such broad reform.

Universal access to early child care has several advantages. Publicly financed systems can provide more equitable access to quality child care and lead to higher labor supply among secondary earners. At the same time, public systems require extensive public funding, which comes at a cost of higher taxes and therefore reduced economic efficiency. Moreover, it is possible that publicly provided child care "crowds out" the private provision of care, with no net increase in child care use or labor supply to the market. Finally, and most importantly, there is an ambig-

¹ The numbers in this paragraph are from the authors' calculations using the Current Population Survey (United States) and Labour Force Survey (Canada) for mothers' labor supply, and the Survey of Income and Program Participation (United States) and National Longitudinal Study of Children and Youth (Canada) for child care use.

² In our data set, which is further described later in the paper, 87.7 percent of the children aged 0–4 who are in some type of nonparental care arrangement are from two-parent families.

uous effect on child outcomes, depending on the implications of time spent in child care versus time spent with the parents.

A full evaluation of publicly financed child care, therefore, requires answers to three questions. First, does public financing affect the quality or quantity of care provided, or does it just lead to a substitution from one form of care to another? Second, if child care use does increase, how large is the associated increase in labor force participation of parents, and what does it suggest about the net cost of the policy (subsidies offered minus new tax revenue collected)? Third, what effect does any change in child care (and associated increases in labor force participation) have on child and family outcomes? Previous studies of child care policy offer at best incomplete answers to one or two of these questions. There has been no evaluation of a full-scale public intervention that can address all three questions.

This paper provides such an evaluation using a major policy innovation in the Canadian province of Quebec in the late 1990s. The Quebec Family Policy began in 1997 with the extension of full-time kindergarten to all 5-year-olds and the provision of child care at an out-of-pocket price of \$5.00 per day to all 4-year-olds. This \$5.00 per day policy was extended to all 3-year-olds in 1998, all 2-year-olds in 1999, and finally all children aged less than 2 in 2000. This dramatic policy change in one of Canada's largest provinces provides a promising research design for evaluating the effect of publicly financed child care.

Our analysis is based on the National Longitudinal Survey of Children and Youth (NLSCY). The NLSCY is an ongoing panel data set that follows the progress of a large, nationally representative sample of Canadian children. We measure the impact of the policy change on mothers' labor supply, child care utilization, and child and parent outcomes. Because concurrent program reforms complicate the inference for single mothers, we focus on married (and cohabiting) women and their children.

The main results of our analysis can be seen in figure 1. We graph measures of mothers' use of nonparental care, their employment, and an index of their children's (aged 2–3) level of anxiety (described later), by wave of the NLSCY, separately for Quebec and the rest of Canada. The Quebec Family Policy was phased in starting in wave 3. In each figure we see good coherence between the time series in Quebec and the rest of Canada in waves 1 and 2, a distinct trend break in Quebec starting in wave 3, and finally a striking change in the relative magnitudes of the measures between Quebec and the rest of Canada in waves 4 and 5 compared to their wave 1 and 2 levels.

Consistent with this evidence, we find that the introduction of universal child care subsidies in Quebec led to a very large increase in the use of care. The proportion of 0–4-year-olds in care rose by 14 per-

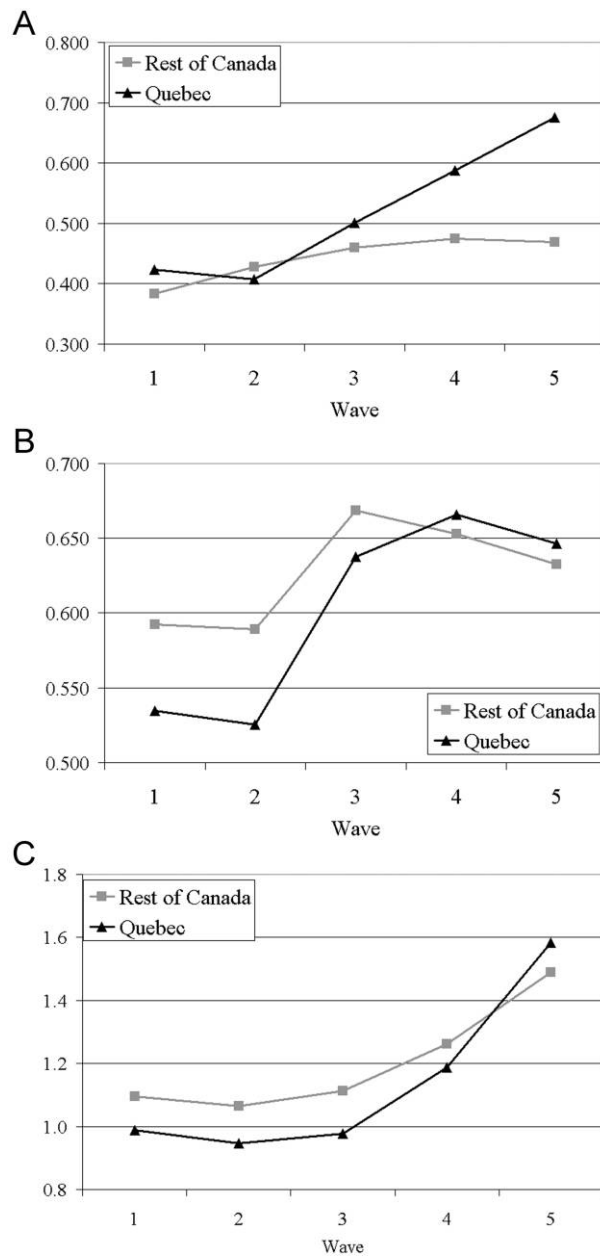


FIG. 1.—Impact of \$5.00 per day program on selected outcomes: A, in any nonparental care, ages 0–4; B, mother works, children aged 0–4; C, anxiety score, ages 2–3. Data are drawn from the National Longitudinal Study of Children and Youth, waves 1–5.

centage points in Quebec relative to the rest of the country, or roughly one-third of the baseline rate. This rise in child care was associated with a sizable increase in the employment of women in two-parent families. Employment rose by 7.7 percentage points in Quebec, or about 14.5 percent of the baseline. The difference between the rise in employment and the rise in child care utilization primarily reflects reduced use of informal child care arrangements, or the “crowding out” of informal child care by this new subsidized child care. Partly as a result of this large crowding out, the taxes generated by the new maternal labor supply fall short of paying for the costs of the increased child care subsidies.

Finally, we also find consistent and robust evidence of negative effects of the policy change on a variety of child outcomes, parenting, and parent outcomes. Child outcomes are worse for a variety of parent-reported measures, such as anxiety, aggressiveness, motor and social skills, child health status, and illness. Measures of parenting and family function are also negatively affected, and there is some evidence of deterioration in parental health and a reduction in parental relationship quality. To our knowledge, effects on parenting and family outcomes have not been investigated previously. Therefore, the results suggest that in this instance more access to child care was bad for children and parents in the dimensions captured in our data. There are, however, interpretations of these findings that are more benign. While some of these explanations appear inconsistent with the data, we cannot rule out the possibility that our findings represent a short-term adjustment to child care, and not a long-run negative impact.

I. Previous Research on Child Care

We have argued that to understand the effect of universal child care subsidies, we need to answer three questions. In this section we review the answers provided in previous research; for a more detailed review, see Baker, Gruber, and Milligan (2005).

The first important question is how the intensity of use of child care depends on its price. If parents are price insensitive or substitute from unsubsidized to subsidized modes of care, then the policy may have no effect on intensity of use. The estimates of own- and cross-price elasticities for child care in the literature vary widely, ranging from quite small (-0.34 in Blau and Hagy [1998]) to quite large (-1.0 or larger in Cleveland, Gunderson, and Hyatt [1996], Powell [2002], and Connelly and Kimmel [2003]).

The second question is, To what extent do child care subsidies influence the labor supply decisions of families? Anderson and Levine (2000) and Blau (2003) review the sizable empirical literature that estimates

the response of labor supply (typically female) to child care costs using U.S. data. Blau reports a wide range of estimated responses of labor supply with respect to child care costs from these studies, although the results from studies using arguably more exogenous variation in child care costs tend toward the lower end of this range.³

These literatures may suffer from an important identification problem: the factors that drive child care costs (number of children or local wages) independently affect maternal labor supply. In response, some recent studies have tried to isolate exogenous variation in child care costs to identify labor supply responses. The estimated elasticities from these studies (e.g., Berger and Black 1992; Gelbach 2002; Herbst 2008; Cascio 2009) are mostly at the lower end of the range of estimates reported by Blau (2003).

The final question of importance for evaluating child care subsidies is how child care utilization affects child development. Parents who respond to subsidies by moving their children from home to subsidized child care are substituting the care of others for their own care. Whether this substitution results in an increase or decrease in the net resources children receive depends on the quality of parental and nonparental child care at stake. If the child care is utilized to permit employment, there may be additional effects on child well-being: more income may have positive effects, but factors such as more parental stress and shorter periods of breast-feeding may have negative effects.⁴ Finally, for those families that already had their children in care, this policy represents a pure income transfer, which could improve child outcomes.

In the child development field, much research in the last decade has been based on a panel study by the National Institute of Child Health and Development—Early Childcare Research Network. An emerging body of evidence suggests some negative behavioral consequences of child care for children. For example, NICHD-ECCRN (2003) finds evidence that the amount of time through the first 4.5 years of life that a child spends away from his or her mother is a predictor of assertiveness, disobedience, and aggression. These negative social outcomes are echoed in research by Belsky (2001), Loeb et al. (2007), and James-Burdumy, Dynarski, and Deke (2008).

Beyond the effect on behavior, a vast body of research studies cognitive and development outcomes. NICHD-ECCRN (2004) associates child care with improved first-grade performance although continued higher

³ Measured elasticities range from 0 to -1.26 , although with the nonlinearity of budget sets it may be hard to compare elasticities across different contexts. The range in the corresponding Canadian studies is -0.156 to -0.388 (Cleveland et al. 1996; Powell 1997; Michalopoulos and Robins 2000; Lefebvre and Merrigan 2008).

⁴ Haider, Jacknowitz, and Schoeni (2003) and Baker and Milligan (2008) provide evidence of how employment after birth displaces breast-feeding.

levels of behavioral problems. This finding echoes research on prekindergarten programs, which has found strong indications of positive cognitive impact.⁵ For example, the recent findings of Magnuson, Ruhm, and Waldfogel (2007) suggest that prekindergarten participation raises cognitive development but sets back behavioral indicators. Previous Canadian studies (Gagne 2002; Lefebvre and Merrigan 2002) find no relationship between the mode of child care and development indicators.

Economic research has also focused on the effects of maternal employment on child development. Ruhm (2004) argues that a consensus view is that the net effects are small: positive effects of employment in the second and third years offset negative effects in the first year.

II. The Policy Change

Our empirical strategy is informed by the details of both the Quebec Family Policy and concurrent changes to the policy environment of families in Quebec and the rest of Canada. The Family Policy was a result of a lengthy public discourse (Jensen 2004) and displays an affinity to the child care programs in European countries (Tougas 2002). We have found little evidence that it emerged in response to any contemporaneous development in Quebec or the rest of Canada, which buttresses our assumption that the reform is exogenous to the outcomes we consider.

Quebec's Universal Day Care Plan

In 1997, the government of Quebec introduced a new set of family policies, including large changes to government subsidies for child care. The centerpiece was a child care program to provide regulated child care spaces to all children aged 0–4 in Quebec at a parental contribution of \$5.00 per day.⁶ Children were eligible whether or not the parents were working. The program was phased in, starting with 4-year-olds in September 1997. Subsequently, 3-year-olds became eligible in September 1998, 2-year-olds in 1999, and children aged 0 and 1 in September 2000.

Subsidized spaces for preschool-age children came in two forms. First, *centres de la petite enfance* (centers for young children—known by the acronym CPE) were created out of existing nonprofit child care centers to serve as organization nodes of the new system. They typically attracted children older than age 2. Younger children were usually placed in

⁵ The literature on Head Start and other early child intervention programs is reviewed by Karoly et al. (1998) and Blau (2003).

⁶ In 2004, the price was increased from \$5.00 to \$7.00 per day. Since this change occurred after the time period covered by our data, we refer to the original \$5.00 amount in this paper.

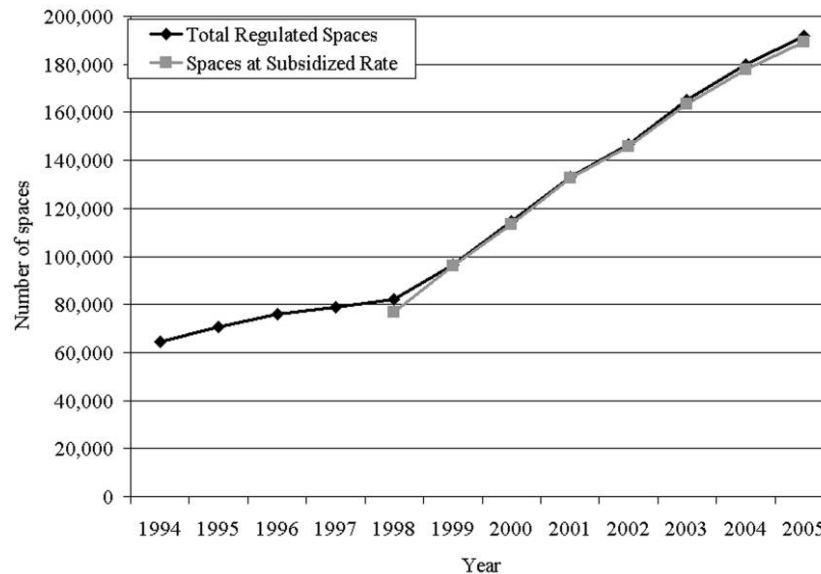


FIG. 2.—Regulated and subsidized spaces in Quebec. The number of spaces is for March 31 in the indicated year. This figure is adapted from table 2 in Lefebvre and Merrigan (2008) and Quebec government statistics (for 2005).

home-based care with a regulated provider. These home-based providers organized as a network affiliated with a CPE. For the balance of the paper, we refer to the new subsidized center-based care and the affiliated home-based care as the “CPE program.” Beyond preschool, the Family Policy introduced (voluntary) full-time kindergarten and subsidized after-school care for children aged 5–12.

The transition to the new system created frictions. Even though the introduction was staggered by age, demand exceeded supply, leading to some queues.⁷ The response to this excess demand was to create new subsidized child care places. In figure 2 we graph the total number of regulated places in the province as of March 31 in each year and starting in 1998 the number of these places available at the subsidized rate. The number of spaces more than doubles between 1997, the last year before the new policy comes into effect, and 2005. The expansion was accomplished by quadrupling the number of places in home-based child care and doubling the number in CPEs. Note, however, that the growth does

⁷ Excess demand is hard to gauge since documented estimates of the queues are unavailable. Waiting lists may include children already in a subsidized spot but wanting to change centers, duplicate counts of children on multiple waiting lists, or children during the transition period who are not yet eligible. Media reports suggest that the queue may be as high as 35,000 children (CBC News Online 2005).

not begin in earnest until 1999; the annual increase in 1998 was only about 4 percent.

The Family Policy also emphasized an increase in the quality of care, in part through regulatory changes. Formal qualifications were raised for both CPE and home-based caregivers.⁸ The government has also implemented new wage policies in the sector to make it a more attractive profession: phased in over a 4-year period starting in 1999, they resulted in an average wage increase of 38–40 percent (Tougas 2002). Changes in operational parameters affecting our age range of interest (0–4) likely had a positive or ambiguous effect on quality.⁹ An audit study based on detailed site visits by Japel, Tremblay, and Côté (2005) indicates that quality in the public child care centers exceeded quality in private options, but still only 61 percent of the public facilities met their criteria for minimum quality. Later in the paper, we offer some direct evidence that suggests that the quality of child care in Quebec along measured dimensions was not falling relative to the rest of Canada over this period.

Effects on the Price of Child Care

Prior to 1997 (and for families without subsidized places after 1997), direct subsidies for child care were available for low-income families in Quebec. There were also refundable tax credits at a rate that depended on family income: from 75 percent for those with the lowest incomes down to 26 percent for those with family incomes greater than \$48,000. These policies provided lower-income families with a substantial subsidy before the introduction of the CPE program.¹⁰ Under the new Family Policy, children in a \$5.00 place are not eligible for any further direct subsidy. In addition, the \$5.00 parental contribution is not eligible for the provincial tax credit for child care expenses, although it remains eligible for a federal deduction. The fact that the pre-1997 child care policies targeted low-income families implies that the effect of the Family Policy on the effective price of child care varies with family income, with little gain for the lowest-income families but larger subsidies at higher incomes.

⁸ Two-thirds of staff must have a college diploma or university degree in early childhood education; previously the standard had been one-third. The government now provides financial support for child care providers who are enrolled in college-level courses in early childhood education. Home-based child care providers affiliated with a CPE faced increased training (24–45 hours) and annual professional development (6 hours) requirements.

⁹ Maximum facility size was raised from 60 to 80 places. Staff/child ratios remained unchanged, with the exception of 4- and 5-year-olds, whose ratios were raised from 1 : 8 to 1 : 10. Parental involvement in the board of directors was raised from 51 percent of members to two-thirds of members. The duration of center operating permits was lengthened from 2 to 3 years.

¹⁰ A full description of these tax subsidies to child care is provided in Baker et al. (2005).

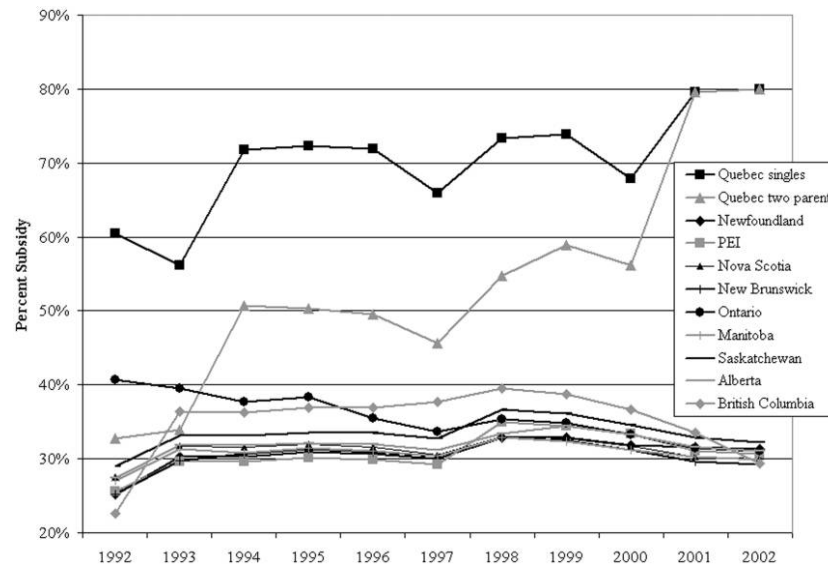


FIG. 3.—Percent subsidy by province. Each data point represents a province-year mean of the percent subsidy variable over the families in the simulation sample. For all provinces, the subsidy rate for two-parent families is shown as well as the subsidy rate for Quebec singles.

In figure 3 we graph the effective subsidy of child care prices by province over the 1990s. We show the rates for two-parent families of children aged 0–4 for all provinces, as well as for single parents in Quebec. The subsidies are calculated as the sum of the direct expenditure subsidy (the direct reduction in child care price from either low-income subsidies or the \$5.00 policy) and the tax subsidy (the tax deduction/credit for any remaining out-of-pocket costs). Subsidies in each province vary with family characteristics such as number of children, one- or two-parent status, and income, so we compute the subsidies for the same set of families across all provinces; as a result, any variation across provinces captures only the differences in legislative environment. More information on the construction of this subsidy variable is provided in Baker et al. (2005).

At the beginning of the time period, Ontario and Quebec have the highest subsidy rates to child care expenditures by two-parent families, and the variation among the remaining provinces is minimal. In 1994 the subsidy rate in Quebec jumps above those in other provinces as a generous refundable tax credit was put in place and then remains steady until the start of the Family Policy. With the introduction of the policy in 1997 for 4-year-olds, the subsidy in Quebec begins an upward trajectory until it settles at almost three times the rate in other provinces by

2002, when all age 0–4 children are eligible.¹¹ This steep rise is in contrast to a relatively constant subsidy rate for two-parent families in other provinces over this time period.

For singles in Quebec, the impact of the Family Policy is only about one-half as large, since single women typically qualified for substantial subsidies before the CPE program was introduced. This finding and contemporaneous policy changes affecting single-parent families lead us to focus our analysis on two-parent families.

In addition to the introduction of the universal CPE child care plan, there were several other changes to family/child benefits in Quebec and the rest of Canada in the late 1990s.¹² In Baker et al. (2005), we carefully model the impact of these changes. For two-parent families, we find little differential impact across provinces, other than the Family Policy in Quebec. Benefits for single women, however, show much more provincial variation, including a sharp increase for lower-income single mothers in Quebec starting in 1998 as a result of a new family allowance targeting low-income parents. Moreover, several provinces implemented welfare reforms through this time period. While two-parent families are eligible for welfare programs in Canada, the proportion that participates is much smaller than for one-parent families. This provides an additional rationale for focusing on two-parent families in the analysis below.

III. Data and Empirical Strategy

Data

Our primary data set for this analysis is the NLSCY, a nationally representative panel survey that follows cohorts of Canadian children, some from as early as birth. The survey is conducted biannually, and we use the 1994–95, 1996–97, 1998–99, 2000–2001, and 2002–3 waves. The initial target population for the NLSCY survey was 0–11-year-olds in 1994. This initial cohort is followed longitudinally across all five waves. Additionally, new age 0–2 cohorts of children have been added to each wave, providing full cross-sectional coverage of ages 0–5. In the fifth wave, the sample includes children aged 0–5 in the cross-sectional component and also the original longitudinal cohort, who are aged 8–19. For this wave, ages 6–7 are missing, which restricts us from a more

¹¹ The jump is not immediate because the phase-in of the new program meant that only 4-year-old children were initially eligible for the \$5.00 program, whereas our simulation sample includes families with children aged 0–4.

¹² See Milligan and Stabile (2007) for details on child benefit and welfare changes of the late 1990s. At that time, 43.5 percent of single mothers had social assistance (welfare) income, whereas only 4.4 percent of married mothers did. Similarly, 74.7 percent of single mothers received the National Child Benefit Supplement, whereas only 17.8 percent of married mothers did.

complete study of cognitive outcomes and other survey questions asked only of school-aged children.

The content of the NLSCY is deep. The data set provides information on a rich set of child care choices as well as tracking parental and teacher evaluations of children's development, test scores, and class rankings. The sample size averages around 2,000 children at each age per year, although some provinces and age groups were oversampled in some waves. For this reason, we use the provided weights in all the results presented here. Our primary sample consists of children aged 0–4, although for some robustness checks we also make use of children aged 8–11. We exclude 5-year-olds to isolate the effect of the child care program from the effect of the expanded full-day kindergarten. The main sample restriction is to include only children from two-parent families, for the reasons outlined earlier.

Empirical Strategy

We estimate difference-in-differences models comparing the outcomes in Quebec and the rest of Canada around the time of this reform. We denote the “prereform” period as waves 1 and 2 of the NLSCY, covering the period 1994–95 to 1996–97. The “postreform” period is waves 4 and 5 of the NLSCY, from 2000–2001 and 2002–3. For outcome variables such as child care use, labor supply, or parent and child outcomes, the generic estimating equation at the individual level is

$$\text{Outcome}_{ipt} = \text{Policy}_p\alpha + \text{PROV}_p\beta + \text{YEAR}_t\phi + X_{ipt}\lambda + \varepsilon_{ipt}, \quad (1)$$

where i indexes individuals, p indexes provinces, and t indexes years. We include year and province dummies, along with a set of control variables X_{ipt} for the parents' characteristics (education level, age group, and immigrant status), size of urban area, number of siblings, and the age and sex of the child.¹³ The primary policy variable is a dummy for being eligible for the CPE program, meaning that the child resides in Quebec and is of an age eligible for a subsidized space in the given year. Since we control for fixed effects for each province and each year, the effect of the child care policy in Quebec is identified by the change in Quebec, relative to other provinces, in 2000 or later relative to 1997 or earlier. For the child care use and labor supply regressions, we also try parameterizing the policy change using the percent subsidy variable

¹³ The parental education groups we define are high school dropout, high school graduate, some post-high school, and university degree. The age groups are in 5-year sets, starting with 16–20 and ending with 46–99. The urban area dummies are for five levels: rural, under 30,000, 30,000–99,999, 100,000–499,999, and 500,000 plus. Siblings are controlled for with dummies for the number of younger siblings (zero, one, two, or more) and another set for the number of same-age or older siblings (zero, one, two, or more).

graphed earlier. We run a battery of robustness checks, outlined below, to increase the plausibility of inferences made from this framework.

Our procedure produces intention-to-treat effects, as we estimate the reduced-form effects on all children rather than only families that choose child care. This carries the advantage of potentially capturing the full impact of the program on both subsidized and unsubsidized care arrangements as well as any peer externalities. Moreover, because of some uncertainty (discussed later) regarding the responses to the child care questions in the survey, our approach allows some flexibility in deciding by which probability of treatment the intention-to-treat effects should be scaled in order to arrive at the impact of treatment on the treated.

We exclude the third (1998–99) wave of the data from the analysis. The phase-in of the program across age groups appears to be an attractive basis for inference, but the lag in increasing the supply of subsidized spaces (see fig. 2) means that there was little scope for an increase in day care uptake in the first years of the program.¹⁴

A disadvantage of our identification strategy is that any Quebec-specific shocks coincident with the Family Policy will bias our estimates. We attempt to address this concern in a number of ways. First, as in figure 1, we enhance our regression results with compelling graphical evidence of how our outcome measures deviate in Quebec and the rest of Canada with the advent of the new policy. The availability of data for two waves before the policy change for most of our variables allows us to demonstrate that we are not just picking up long-running trends in differences between Quebec and the rest of Canada. Second, we run a variety of falsification checks on the results, such as examining effects among 8–11-year-olds in Quebec (who were less affected by the policy) and controlling for contemporaneous changes in economic conditions. These specification checks are presented after our central results.

Finally, an important issue with this type of difference-in-differences analysis is the correct computation of the standard errors. To account for the fact that the policy variation we use is at the province-year level, we report robust standard errors clustered on province-year cells to account for any dependence of the errors within province-year groups. This approach does not address any temporal correlation in the errors or the fact that we have a small number of treatment provinces. However, we provide evidence that our inferences are robust to approaches that address these issues.

¹⁴ Estimates from a triple-difference regression including all waves indicate a zero impact on use and offsetting changes in composition: movement toward center-based care (8 percentage points) and away from own home care (3 points) and other home care (5 points).

TABLE 1
DESCRIPTIVE STATISTICS

DEMOGRAPHICS/CONTROL VARIABLE	QUEBEC		REST OF CANADA	
	Waves 1–2	Waves 4–5	Waves 1–2	Waves 4–5
Big urban	.579 (.494)	.574 (.495)	.428 (.495)	.458 (.498)
Older siblings	.715 (.739)	.713 (.718)	.796 (.761)	.759 (.731)
Younger siblings	.268 (.488)	.236 (.454)	.255 (.476)	.253 (.476)
Mother age	30.9 (4.9)	31.2 (5.3)	31.7 (5.1)	32.4 (5.4)
Mother immigrant	.089 (.285)	.114 (.318)	.214 (.410)	.240 (.427)
Mother high school dropout	.133 (.340)	.115 (.320)	.106 (.308)	.090 (.286)
Mother university degree	.203 (.402)	.280 (.449)	.206 (.404)	.280 (.449)
Father age	33.5 (5.4)	34.0 (5.8)	34.1 (5.7)	34.9 (6.0)
Father immigrant	.097 (.296)	.121 (.326)	.208 (.406)	.236 (.425)
Father high school dropout	.168 (.374)	.157 (.364)	.138 (.345)	.110 (.312)
Father university degree	.194 (.395)	.239 (.426)	.214 (.410)	.262 (.440)
Boy	.509 (.500)	.518 (.500)	.509 (.500)	.513 (.500)
Child age	2.03 1.42	2.05 1.41	1.99 1.42	2.03 1.42
In care	.415 (.493)	.629 (.483)	.405 (.491)	.472 (.499)
Mother works	.530 (.499)	.657 (.475)	.591 (.492)	.643 (.479)
Subsidy rate	.502 (.000)	.795 (.000)	.349 (.025)	.322 (.019)

NOTE.—Displayed are the means for each variable, with the standard deviation beneath in parentheses. The data come from the National Longitudinal Study of Children and Youth. The sample is split into Quebec and the rest of Canada, for wave 1 (1994–95), wave 2 (1996–97), wave 4 (2000–2001), and wave 5 (2002–3). The sample contains 34,042 observations for children aged 0–4.

Table 1 shows the means for our control variables and some key dependent variables, divided into Quebec and the rest of Canada, before and after the policy change. There are no noticeable differential trends in Quebec, relative to the rest of the nation, except for impacts on use of child care, subsidy rates to child care, and maternal labor supply.

IV. Child Care Use and Labor Supply

Child Care Use

The initial NLSCY question about child care use is asked of parents of children aged 0–11.¹⁵ This question acts as a “gateway,” since no further child care questions are asked for those who respond in the negative. Those who answer in the positive are asked a series of questions about the modes of child care used and for how many hours each is used. We classify the child care modes into (a) institutional care (mostly day care centers), (b) care in the home, and (c) care outside the home. For care in the child’s or another’s home, we observe whether it is provided by a relative and whether the caregiver is licensed. Because the care subsidized by the Family Policy can be provided both through CPEs and through licensed home-based providers, our focus is on institutional care and licensed care outside the home.

Table 2 presents the results for our child care use variables. Each row shows the coefficient of interest from two separate regressions. The first is a difference-in-difference estimate using the dummy eligibility indicator (ELIG) for observations in Quebec in waves 4 and 5, and the second is the result from a model including the province-year subsidy rate. We also report the pre-Family Policy mean of each dependent variable in Quebec and its standard deviation. The first row shows that the odds a child was in child care rose by 14.6 percentage points in Quebec, relative to the remainder of Canada, after this policy change. This is a very sizable increase that amounts to more than a third of the baseline rate of child care utilization. As noted earlier in figure 1A, there is a clear trend break in the use of child care in Quebec at the time the policy was introduced. In regressions not shown here, we find that the increase in use is almost entirely in full-time (more than 20 hours) care.¹⁶

The next rows in table 2 show estimates of changes in the various types of child care arrangements. There is a very large rise in institutional care that is essentially equal to the overall rise in child care. There is no change in care in own home or in other’s home. This is puzzling given the increase in home-based care that was part of the CPE program. The next three rows resolve the mystery by disaggregating the care in other’s home. There was a shift from care provided by relatives and nonlicensed nonrelatives to care provided by licensed nonrelatives. Li-

¹⁵ The wording of the question is, Do you currently use child care such as day care, babysitting, care by a relative or other caregiver, or a nursery school while you (and your spouse/partner) are at work or studying?

¹⁶ Datta Gupta and Simonsen (2008) provide Danish evidence that children who had been exposed to longer weekly durations in child care at age 3 showed significantly worse behavioral outcomes at age 7.

TABLE 2
LABOR SUPPLY AND CHILD CARE RESULTS

Dependent Variable	Observations	Mean (Standard Deviation)	ELIG Dummy	Percent Subsidy
In any care	33,864	.415 (.493)	.146*** (.033)	.458*** (.089)
Institutional care	33,864	.110 (.313)	.152*** (.030)	.474*** (.077)
Care in own home	33,864	.072 (.258)	-.009 (.008)	-.023 (.026)
Care in other's home	33,864	.230 (.421)	.002 (.017)	.003 (.051)
Breakdown of care in other's home:				
Licensed nonrelative	33,864	.044 (.206)	.048*** (.004)	.142*** (.012)
Nonlicensed nonrelative	33,864	.135 (.342)	-.025** (.011)	-.072** (.032)
Care in relative's home	33,864	.051 (.219)	-.021*** (.007)	-.067*** (.021)
Mother works	33,788	.530 (.499)	.077*** (.006)	.242*** (.016)
Mother works/child in care	33,634	.356 (.479)	.125*** (.016)	.384*** (.045)
Mother works/child not in care	33,634	.173 (.379)	-.048*** (.015)	-.142*** (.047)
Mother doesn't work/child in care	33,634	.059 (.236)	.023 (.018)	.077 (.053)
Mother doesn't work/child not in care	33,634	.412 (.492)	-.100*** (.020)	-.320*** (.049)

NOTE.—The sample contains all 0–4-year-old children in waves 1, 2, 4, and 5 with valid responses. Each row represents a different dependent variable. For each dependent variable we show the number of observations, the preprogram mean for Quebec (with standard deviation in parentheses), the coefficients from separate regressions on the eligibility dummy, and the percent subsidy variable. Also included in the regressions is a set of control variables including dummies for the child's age, sex, number of older siblings, and number of younger siblings; the mother's age, education, and immigrant status; the father's age, education, and immigrant status; and the size of the urban area and the province in which the family lives, as well as dummies for each wave.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

censed nonrelatives would include the home-based care that is part of the CPE program. Clearly, this policy change had major effects on the use of child care.

Labor Supply

The remainder of table 2 reports the effects of this policy on the labor supply of women in two-parent families.¹⁷ There is a rise in the em-

¹⁷ The unit of observation for the analysis in the paper is the child. To maintain consistency, we also use this to analyze mothers' labor supply, although a woman with more than one child aged 0–4 may appear in the sample multiple times. We have tried restricting the sample to one observation per mother and found little difference.

ployment of women in Quebec, relative to the rest of Canada, of 7.7 percentage points, or 14.5 percent of baseline participation. This is consistent with the graphical evidence shown earlier in figure 1*B*. In regressions not shown here, we find that the increase in work was dominated by women working more than 30 but less than 40 hours per week. The labor supply results are qualitatively similar to those found by Lefebvre and Merrigan (2008), who explore the labor market effects of the Family Policy using different data and a different sample.

It is notable that the impact of the program on labor supply is only about half as large as the impact of the program on child care utilization. There are two possible explanations for this finding. First, many women may be using child care without working. The other possibility is a change in reporting: some women may have been using informal child care and answered “no” to the child care usage question, but as they switched to the formal sector they report their care.

The final four rows of table 2 investigate these two hypotheses by dividing the sample of mothers into four groups: working and using child care; working and using no child care (presumably a mismeasurement that is proxying the informal care of these 0–4-year-old children); not working and using child care; and not working and using no child care. The first two rows show that there is a reduction of 4.8 percentage points in the share of women who report working with no child care for their small children, and the share who report working with child care rises by 12.5 percentage points (the sum of these is the 7.7-percentage-point increase in work). Thus, roughly one-third of the 14.6-percentage-point rise in child care use reported in the first row of table 2 appears to be a shift from unreported informal care to more formal care (crowding out of informal care) rather than a net increase in child care use. Finally, the third row shows that there is a small increase in the share of women who use child care but do not work.

These findings frame our investigation of child outcomes. The Quebec Family Policy is not purely an instrument for increased child care use and labor supply. Rather, the effect of the policy is a mix of increased labor supply with child care use, a shift in the mode of child care (from informal to formal care), and a small rise in child care use without increased labor supply.

The estimates for child care use and labor supply can be used to estimate the net budgetary cost of the program, accounting for offsetting tax revenues from increased maternal labor supply. We provide such an estimate in Baker et al. (2005) and conclude that 40 percent of the costs of the child care subsidy are covered by the income and payroll taxes on the extra labor the subsidy encourages. That is, the large shift from informal (unsubsidized) to formal (subsidized) child care, as well as some increased child care use by mothers who do not work, resulted

in a significant net cost of this program, despite the large rise in labor supply.

V. Child Outcomes, Parenting, and Parent Outcomes

Child Outcomes

As discussed in Section I, there is some controversy over the effects of maternal work and child care use on child outcomes. The policy change in Quebec, along with the rich data on child outcomes available in the NLSCY, offers an opportunity to address this important question more completely. Given the ages of the preschool children in our sample, we do not observe many of the school performance or test score measures that are familiar to economists in this context. Fortunately, there is enormous expertise in other disciplines measuring behavioral and pre-cognitive outcomes of young children, and this expertise is incorporated into child outcome measures available in the NLSCY.¹⁸

We consider several summary scores of the behavioral questions in the NLSCY: scores for hyperactivity-inattention, general anxiety, separation anxiety, and physical aggressiveness and opposition.¹⁹ In online Appendix B, we provide estimates for the individual questions that make up these scores using the same specification as we use for the scores. We also use a measure of motor and social development and the one test score available for our age range—the Peabody Picture Vocabulary Test (PPVT). Finally, we construct five measures of health, including indicators for excellent health, for the child never having had a nose/throat or an ear infection, for an asthma attack in the last 12 months, and for an injury in the last 12 months.

Table 3 presents the results for these child outcomes from the difference-in-differences specification. The first four rows show results for our behavioral indices for children aged 2–3. In all cases the estimates indicate that behavior deteriorated in Quebec, relative to the rest of Canada, over this period. Both the emotional disorder–anxiety score and the physical aggression–opposition score are significant at conventional levels. The standardized motor and social development score for children aged 0–3 shows a significant decline of -1.647 , which is more than 10 percent of a standard deviation. The change in the PPVT score is not significant, but the small number of age 4 observations contributes to the imprecision of this estimate. The 95 percent confidence interval

¹⁸ A detailed report on the measures in the NLSCY that we use is available in online App. B and from the authors on request.

¹⁹ It is important to keep in mind that these are not independent measures. For these five child measures, the average of the 10 pairwise correlations is 0.260, and only two exceed 0.4. For the child health measures, the average correlation is 0.077, and for the parenting measures (discussed below), the average correlation is 0.244.

TABLE 3
RESULTS ON CHILD BEHAVIOR, FAMILY, AND HEALTH

Dependent Variable	Observations	Range	Mean (Standard Deviation)	ELIG Dummy
A. Child Behavior and Health				
Hyperactivity-inattention, ages 2–3	14,494	0–14	4.102 (3.134)	.103 (.146)
Emotional disorder–anxiety score, ages 2–3	14,555	0–12	.967 (1.343)	.120** (.055)
Separation anxiety score, ages 2–3	14,580	0–10	2.668 (2.029)	.099 (.085)
Physical aggression and opposition, ages 2–3	14,435	0–16	4.375 (3.041)	.380*** (.084)
Standardized motor and social development score, ages 0–3	26,176	1–177	99.32 (15.03)	–1.65*** (.46)
PPVT score, scaled, age 4	5,210	45–160	99.76 (15.15)	.36 (.75)
In general, child is in excellent health, ages 0–4	33,891	0/1	.641 (.480)	–.055*** (.016)
Child never has nose or throat infections, ages 0–2	28,175	0/1	.404 (.491)	–.140*** (.025)
Child has never had an ear infection, ages 0–2	28,161	0/1	.438 (.496)	–.057*** (.019)
Had asthma attack in past 12 months, ages 0–4	33,867	0/1	.955 (.208)	–.003 (.004)
Child has been injured in past 12 months, ages 0–4	33,878	0/1	.071 (.258)	.006 (.008)
B. Parent Behavior and Health				
Hostile, ineffective parenting, ages 2–4	20,017	0–25	8.320 (3.842)	.728*** (.091)
Consistency, ages 2–4	19,809	0–20	14.048 (3.266)	–.504*** (.117)
Aversive parenting, ages 2–4	20,116	0–20	8.346 (2.014)	.198*** (.067)
Family functioning, ages 0–4	33,248	0–36	7.188 (4.979)	.257 (.173)
Mother health status is excellent, ages 0–4	33,708	0/1	.406 (.491)	–.011 (.011)
Father health status is excellent, ages 0–4	33,586	0/1	.449 (.497)	–.029** (.012)
Mother depression score, ages 0–4	29,595	0–36	4.199 (4.563)	.422*** (.119)
Satisfaction with relationship, ages 0–4	26,473	0–11	9.505 (1.679)	–.194*** (.025)

NOTE.—For each dependent variable we show the number of observations, the preprogram mean for Quebec (with standard deviation in parentheses), the range for the dependent variable, and the coefficient from a regression on the eligibility dummy. Also included in the regressions is a set of control variables including dummies for the child's age, sex, number of older siblings, and number of younger siblings; the mother's age, education, and immigrant status; the father's age, education, and immigrant status; and the size of the urban area and the province in which the family lives, as well as dummies for each wave.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

includes effects as large as 9.7 percent of a standard deviation, which when scaled for treatment (see below) is quite large.²⁰

There are also sizable negative effects on health indicators. There is a significant negative effect on the odds of being in excellent health of 5.5 percentage points. There is a very large and significant 0.140 reduction in the odds of never having had nose and throat infections and a similar effect on ear infections. In contrast, there is no change in asthma. These results suggest an increased risk of exposure to communicable illnesses. Finally, there is no evidence of a change in the odds of a recent injury following the large switch into nonparental care.

Many of these estimated effects appear small relative to the means and standard deviations. However, the estimates we report are intention-to-treat effects and must be divided by the probability of treatment to arrive at the effect of the treatment on the treated. Measuring the effect of the treatment on the treated is quite difficult, however, since defining what it means to be treated is not clear. Treatment could be defined as exposure to maternal labor supply (in which case 7.7 percent of children are treated), the decrease in mothers caring for their children but not working (in which case 10 percent of children are treated), or the increase in self-reported use of care, which may include switching between informal and formal care (in which case 14.6 percent of children are treated). Furthermore, some children may also have experienced an increase in the intensity of child care, because the CPE program was available for up to 12 hours of care per day.

While we are unable to resolve this issue, we can plausibly bound the effects by assuming that between 7.7 percent and 14.6 percent of children are treated. Doing so makes the effects appear quite large. For example, for motor and social skills, the estimates suggest that eligibility for the subsidized child care plan leads to a 1.64 percent decline in skills relative to the mean, for an effect on the treated of between 11.3 percent and 21.4 percent, which is between 75 percent and 133 percent of a standard deviation. The effects are as large for other behavioral measures. For example, the results for anxiety imply a 12.4 percent rise relative to the mean score of 0.967 for the intention-to-treat effect, which is 8.9 percent of a standard deviation. When scaled for treatment, the effect is between 61 percent and 116 percent of a standard deviation. The physical aggression score scales to similar magnitudes. Some of the

²⁰ We also estimated regressions substituting indicators for child care use or institutional care use as the “policy” variable. This simple ordinary least squares (OLS) specification ignores the selection of children into child care but can be compared to the policy-induced variation we do exploit. For the child behavior measures (and most of our other measures), the “naïve” OLS specification shows smaller (often statistically insignificant) effects or impacts of opposite sign consistent with the idea that working mothers with children in care are positively selected.

health effects are also quite large: the rise in the rate of nose/throat infection for all children is 34.7 percent of the prepolicy mean, suggesting a policy effect on the treated of 237–451 percent. Whether we use the upper or lower bound, our estimates imply large effects of maternal work and child care use.

In figure 4, we present graphs of the additional child outcome measures similar to the graph for anxiety in figure 1. Hyperactivity and aggression actually fall in Quebec relative to the rest of Canada before the policy, but there is a sizable relative rise in Quebec between waves 2 and 5. Our indicators for the absence of nose and throat infections are flat in both Quebec and the rest of Canada through the first two waves and then fall dramatically in Quebec in the postpolicy period.

Specification Checks

The results presented thus far show clearly that child care utilization and maternal labor supply rose whereas child outcomes deteriorated in Quebec after this policy change, both relative to other provinces and relative to previous trends. We also showed in table 1 and in the figures that there were no obvious other trends in these variables in Quebec relative to the rest of Canada during this period. We now turn to several specification checks that investigate the robustness of these inferences.

In the first row of table 4 we reproduce the estimates for our child-focused dependent variables and their province-wave clustered robust standard errors. For hyperactivity, anxiety, and aggression we report estimates for the scores for children aged 4 rather than the age 2–3 scores reported in table 3. This is to facilitate comparisons of the results to an older (ages 8–11) comparison group, for whom only the age 4–11 score is available.²¹ A full set of robustness checks for the age 2–3 scores and the other dependent variables is available in online Appendix table A4.

We first explore the robustness of our findings to alternative methods of statistical inference. To account for any serial correlation in the errors, we start by assuming that the dependence is confined to the pre- and postpolicy periods. Clustering on province/pre- and postpolicy period cells leads to the standard error estimates in the next row, which in some cases are substantially smaller.²² The results are very similar when we cluster on province alone (which does not assume a temporal break in the dependence). In the next row we follow the method of Bester, Conley, and Hansen (2008) allowing dependence over time and within

²¹ The behavior questions differ for children aged 2–3 and children aged 4–11.

²² The pattern of statistical significance is similar to our reported results in table 3 if we implement this structure using two-step estimation ending in a 20-observation regression (10 provinces by 2 time periods) with eight degrees of freedom.

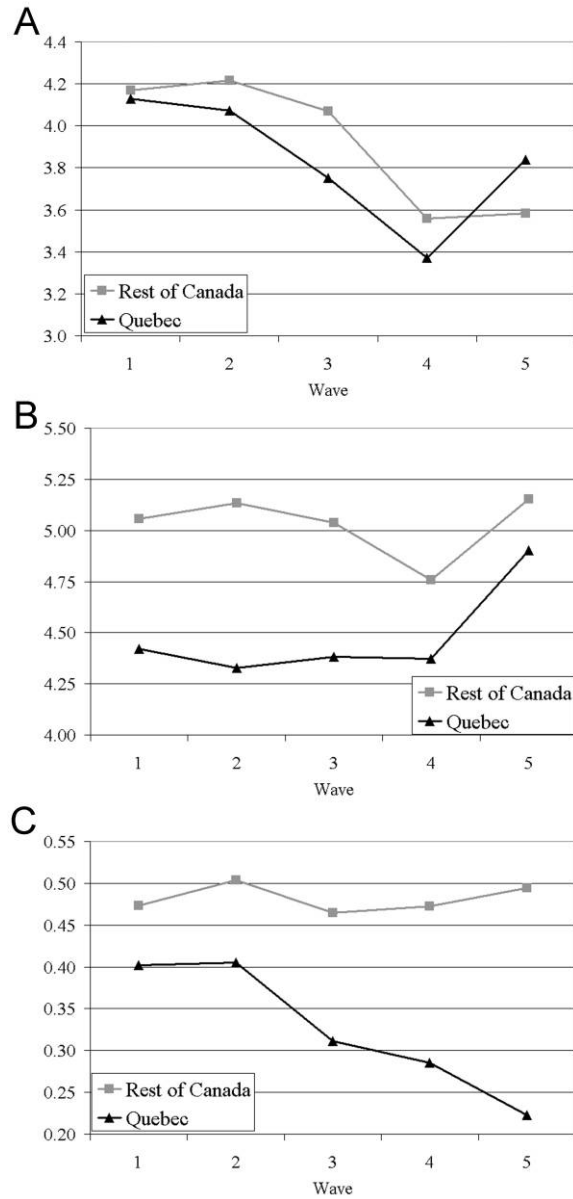


FIG. 4.—Children's outcomes: A, hyperactivity-inattention score, ages 2–3; B, physical aggression and opposition score, ages 2–3; C, child never has nose/throat infections, ages 0–2. Data are drawn from the National Longitudinal Study of Children and Youth, waves 1–5.

regions. We cluster using four regional groups and base inference on a t -distribution with three degrees of freedom.²³ These results are slightly more conservative, but the original pattern of significance persists. Finally, we implement the method of Hansen (2007): feasible generalized least squares (FGLS) assuming an AR(1) specification of the errors. The inferences (and parameter estimates) are very similar to our base case.

To address the fact that we have a small number of treatments, we next report p -values from simulations following the guidance of Anderson and Meyer (2000), Conley and Taber (2005), and Gruber and Hungerman (2008). The reported p -values place the estimate in the first row of the table in the context of a distribution of “placebo” estimates in nontreated provinces.²⁴ By this method, the increase in care in Quebec is far outside anything seen in other provinces, but the increase in mothers’ employment lacks significance. Both hyperactivity and anxiety scores fall in around the 0.10 significance level (one just over and one just under). In contrast, the aggression, motor and social, and child in excellent health results come in with simulated p -values of 0.000, indicating strong significance.

Next we estimate a specification that controls for province-specific economic conditions by adding the province-specific prime-age male unemployment rate. Including male labor market outcomes is not ideal to the extent that there may be spillovers of this policy to male labor supply, but such spillovers are likely to be small.²⁵ These results are shown in the next row of table 4, and there is little difference from our base estimates.

Finally, where possible, we use older children as a comparison group. These children were in the same economic environment, and so we can use them to check for false positives. We use 8–11-year-olds who currently have no younger siblings and were never eligible for subsidized child care when aged 0–4 or extended full-day kindergarten at age 5. This way there is no potential “contamination” of the comparison group from

²³ Their method is appropriate for cases with a small number of groups. We follow the guidance to use geographically large groups (Atlantic provinces, Quebec, Ontario, and the western provinces) of roughly the same number of observations. We also implement the sample size correction on the cluster robust standard errors provided by STATA, as they suggest. Note that this method may not be completely applicable to our case since it makes the assumption that the distribution of regressors is asymptotically the same in each group.

²⁴ We take the nine nontreated provinces and randomly assign each province a “placebo” treatment for the wave 4/5 period with a 0.10 probability. We repeat this simulation 10,000 times to generate a distribution of parameter estimates from which we derive our reported p -values.

²⁵ We tried to explore this issue using measures of male labor supply in the NLSCY and found insignificant effects.

TABLE 4
EXTENDED RESULTS FOR CHILDREN VARIABLES

	In Any Care	Mother Works	Age 4-11 Hyperactivity	Age 4-11 Anxiety	Age 4-11 Aggression	Motor and Social	Never Had Nose Throat Infection	Child in Excellent Health
Base specification	.146 (.033)***	.077 (.006)***	.484 (.194)**	.489 (.134)***	.471 (.127)***	-1.647 (.461)***	-.140 (.025)***	-.055 (.016)***
Province-wave cluster								
Province-pre/post cluster								
Four region cluster (3 df)								
Hansen (2007) FGLS estimation AR(1)	.154 (.010)***	.079 (.011)***	.493 (.113)**	.532 (.056)***	.476 (.108)**	-1.551 (.113)***	-.122 (.011)***	-.052 (.005)***
Simulated <i>p</i> -values for base	.000***	.231 (.007)***	.106 (.177)***	.092* (.105)***	.000*** (.147)***	.000*** (.235)***	.000*** (.015)***	.000*** (.010)***
With unemployment control	.122 (.026)***	.074 (.008)***	.252 (.241)	.310 (.118)***	.417 (.174)**	-1.422 (.534)***	-.136 (.026)***	-.067 (.016)***
Age 8-11, no younger sibling comparison group	-.005 (.022)	.017 (.027)	.089 (.285)	.843 (.145)***	.058 (.086)	. . .	. . .	-.044 (.034)

Ages 8–11, with younger sibling	.119 (.020)***	.140 (.022)***	-.189 (.239)	.003 (.245)	.005 (.135)	. . .	. . .	. . .	-.006 (.024)
Ages 0–2 only	.167 (.040)***	.090 (.013)***	. . .	. . .	. . .	-2.536 (.679)***	-1.134 (.023)***	-1.134 (.023)***	-.049 (.016)***
Ages 3–4 only	.117 (.027)***	.055 (.019)***	.484 (.194)***	.489 (.134)***	.471 (.127)***	.740 (.850)	-.160 (.043)***	-.160 (.043)***	-.066 (.018)***
Just single parents	.153 (.056)***	.040 (.037)	.274 (.620)	1.071 (.313)***	.343 (.435)	2.167 (1.495)	-.169 (.041)***	-.169 (.041)***	-.035 (.039)
High school or less	.083 (.046)*	.030 (.019)	.519 (.355)	.726*** (.170)	.792*** (.177)	-2.405 (.447)***	-.085 (.023)***	-.085 (.023)***	-.102 (.019)***
Some post-high school	.170 (.028)***	.095 (.008)***	.479 (.364)	.380*** (.150)	.326*** (.148)	-1.431 (.599)***	-.158 (.028)***	-.158 (.028)***	-.034 (.017)**

NOTE.—Each row shows the results for the same set of dependent variables but a different method of estimation or inference. The sample includes children aged 0–4 except where noted. The first row shows the base coefficients, followed by three different standard errors. See the text for details. All regressions include a set of control variables for the child's age, sex, number of older siblings, and number of younger siblings; the mother's age, education, and immigrant status; the father's age, education, and immigrant status; and the size of the urban area and the province in which the family lives, as well as dummies for each wave.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

previous exposure to care eligibility.²⁶ These children are still not an ideal comparison group, however, because through the Family Policy they became eligible for subsidized after-school care. This biases the results for the comparison group in the same direction as the treatment group, however, so that comparisons will understate only policy effects. Across this row in table 4, there is only one significant result—for the anxiety score. There is no increase in care or in maternal employment among families with older children in Quebec, in sharp contrast to what was observed for ages 0–4. This supports our causal interpretation of the results for the younger children.

We next use a comparison group of 8–11-year-olds who have a younger sibling. This is potentially an interesting group because while their mothers face an incentive for increased work, the older children themselves would not participate in the full-day CPE program even if their younger siblings did. We observe a large increase in (after-school) care for these children and in their mothers' employment. However, our measures of child well-being show no indication of change at all, suggesting that maternal work had little impact on this set of older children.

Alternative Samples

In the remaining rows of table 4 we explore heterogeneity in our results by looking at some different samples and subsamples. To begin we present results for children aged 0–2 and 3–4 separately. We find strong impacts on labor supply and child care utilization for both groups, although the effects are larger for children aged 0–2. The effect for the excellent health indicator is about the same in each subsample, but for the motor and social score there is a much stronger negative impact for the younger age group. Moreover, in online Appendix table A4 we show that for the age 2–3 behavioral scores, the impact on the younger children is greater. This suggests that the negative effects of increased child care utilization are more evident in this age group.

In the next row are results for single parents. As noted earlier, the Family Policy caused a smaller change in subsidy rates for this group, and they were more likely affected by changes to welfare and child benefit policies. The estimates reveal that the Family Policy had a large impact on the use of nonparental care for this group but a much smaller impact on maternal employment. This indicates that the program induced a large rise in child care use among nonworking single mothers. In some cases the impacts on the behavioral measures are larger for this group than for children in two-parent families (see also App. table

²⁶ There are no children aged 5–7 in Quebec in waves 4–5 who satisfy these conditions, so we use only those children aged 8–11 in the comparison group.

A4). These results suggest that it may be child care use, not maternal labor supply, that drives the negative behavioral findings in our two-parent sample, although the inferences from the singles sample are clouded by the confounding policy changes.

The final sample stratification we consider is by parental education. The Family Policy provided a much larger increase in subsidies for middle- and higher-income two-parent families than for lower-income two-parent families. While income is endogenous to the labor supply response, we can explore incomes indirectly by examining low- versus high-education parents. The impacts of the reform on child care utilization and labor supply are indeed much stronger for the higher-education group. The results on child outcomes are somewhat mixed, however. It is difficult to interpret these differences because there are both differences in uptake across the two education groups and potential heterogeneous impacts of uptake.

Interpretation

Our results are striking in their consistent indication of a substantial negative impact of universal child care subsidies on children in two-parent families. Moreover, both the graphical evidence and the lack of consistent findings for 8–11-year-olds buttress the claim that our results are causally related to the policy change. Nevertheless, this is a reduced-form finding and is subject to numerous interpretations. We discuss these interpretations, and data we can bring to bear on them, in this subsection.

First, it is possible that these findings are simply reporting artifacts. For example, higher exposure to child care could lead to increased reports of bad outcomes with no real underlying deterioration in child behavior, if child care providers identify negative behaviors not noticed (or previously acknowledged) by parents. Alternatively, perhaps parents are more stressed because of increased family labor supply, so they are more likely to report negative child behaviors.²⁷

While we cannot rule out these alternatives, they seem unlikely given the consistency of our findings both across a broad spectrum of indices and across the categories that make up each index (as shown in online App. tables A1–A3). In particular, these alternatives would not suggest such strong findings for health-based measures or for the more objective evaluations that underlie the motor and social skills index (such as counting to 10 or speaking a sentence of three words or more).

Second, these findings may be transitional, reflecting problems Que-

²⁷ We provide an in-depth discussion of potential biases in parental reports in online App. B.

bec experienced as its child care system expanded rapidly—lower child care quality, for example. Using the Canadian Labour Force Survey, we evaluated the earnings, age, and education level of child care workers over time across provinces. We found no evidence that child care workers were becoming less qualified in Quebec over time; if anything, qualifications were rising relative to the rest of Canada.²⁸ In the NLSCY, we find that the median caregiver ratio among children in care in waves 4 and 5 is 5.0 in Quebec and 4.0 in the rest of Canada. However, we cannot compare these to prereform ratios because they are not reported in the NLSCY until wave 3.

If the child outcome results do reflect deterioration in the outcomes of small children, there remains the question of whether they represent short-run problems or have longer-run consequences. A simple mechanism for the former is the initial costs of socialization. If all children have difficulties when they are first in a social environment interacting with other children, then perhaps the CPE policy brought forward this “day of reckoning,” moving up these problems of kindergarten to an earlier age. In that case, the welfare implications of our findings are ambiguous. Indeed, it might be beneficial that this policy exposed children to these costs earlier on, so that they were better prepared for the educational and health consequences of attending school (e.g., they have their infections at age 2 or 3 rather than age 5 or 6).

The best way to distinguish these views would be to examine the impact of the CPE policy on the long-term outcomes of this exposed cohort. Unfortunately, the data to carry out this evaluation are not yet available. Nevertheless, a partial test of this particular socialization explanation is to compare children with and without siblings. Presumably, the socialization “shock” was larger for those without siblings, so this story would imply a larger effect on this group. We have estimated our models separately for children aged 0–4 with and without siblings, finding no consistent evidence of a stronger effect on one group or another. While not ruling out the socialization story, this finding is not consistent with it.

²⁸ We selected workers reporting North American Industrial Classification System code 6244 (child day care services). Comparing changes in average values between 1994–97 and 2000–2002 (to match our NLSCY data), we find that the average age of child care workers changes almost identically (and is almost identical) in Quebec and the rest of Canada; the proportion of workers who work full-time rises more in Quebec (by 5 percentage points); and the proportion of workers with some postsecondary education rises by 26.7 percentage points in Quebec, to reach 73.2 percent, and rises by 21.3 points in the rest of Canada to reach 66 percent. For earnings, we can compare averages between 1996–97 and 2000–2002. Earnings rise in Quebec by 12 percent (nominally) and fall in the rest of Canada by 6 percent.

Parenting and Parent Outcomes

The NLSCY gathers data not only on child outcomes but also on the quality of parental interactions with children and on the well-being of parents themselves. Both of these might plausibly be affected by increased use of child care and labor supply induced by the CPE program. As best we can tell, the effect of nonparental care on these sorts of outcomes has not been investigated in the literature.

We use three measures of the quality of parental interactions provided by the NLSCY: “hostile and ineffective parenting,” “parental consistency,” and “aversive parenting.”²⁹ These indices are aggregated by the NLSCY from parental responses to individual questions. Once again, while parent-reported, these measures have been developed and tested by child development experts.

Panel B of table 3 presents difference-in-differences estimates of the effects of the policy change on these indicators of parental interaction quality. For each of these scales, there is strong evidence of less effective parenting after the new policy was put in place. There is a significant rise in the hostile-ineffective parenting index of about 8.7 percent of its baseline value, a significant decline in the consistent parenting index of about 3.6 percent of its baseline value, and a significant rise in aversive parenting of about 2.4 percent of the baseline. We plot the time trends for the three parenting measures in figure 5. While aversive parenting shows little closing of the Quebec–rest of Canada gap, the other measures show remarkable changes consistent with the parent-child relationship getting worse in Quebec in waves 4 and 5.

The final rows of table 3 consider the effects on measures of parental well-being. The self-assessed health status of the mother and father is available in the NLSCY, along with a depression score for one of the parents. (We present the depression results only for mothers since the mother was the primary respondent in the vast majority of families.) In addition, the survey contains a question about the satisfaction of the survey respondents with their spousal relationship, on a scale of 1–11.

The estimated coefficients indicate deterioration in paternal health around the time of this policy, with a reduction of 2.9 percentage points in the odds that the father reports himself in excellent health. For mothers, the estimated effect on self-reported health is smaller at -0.011 and not statistically significant. For mothers, however, we have striking evidence of an increase in depression: the mother’s depression score is estimated to increase by 0.422, or 9.2 percent of a standard deviation. There is also a very striking negative effect on reported relationship

²⁹ We do not include the available “positive interactions” variable because it is mechanically related to time exposure to children and so is not very indicative of a deteriorating relationship conditional on increased child care.

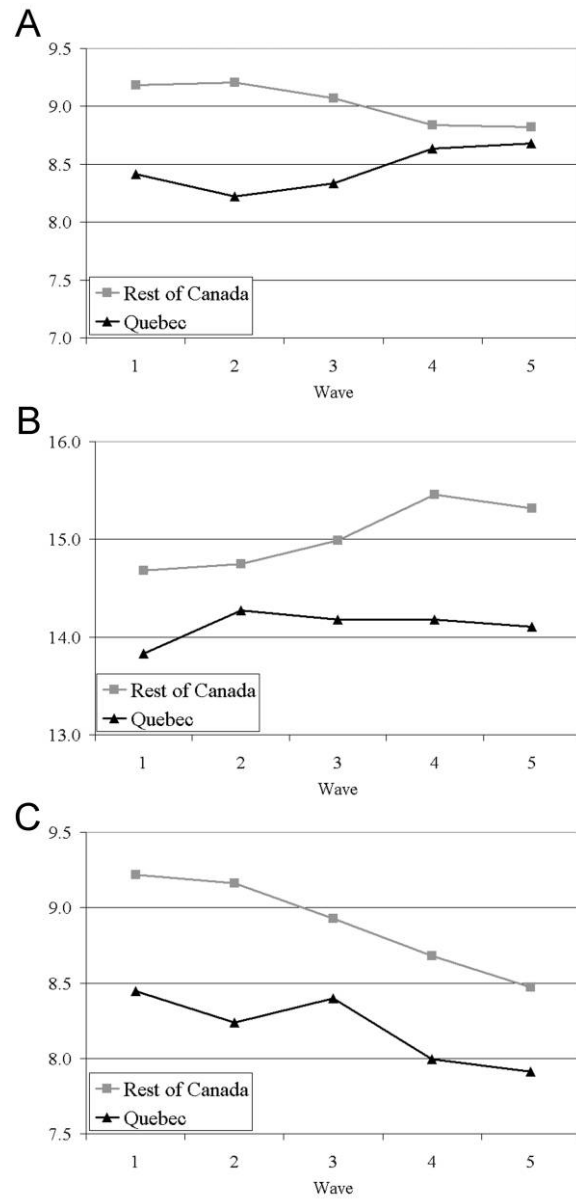


FIG. 5.—Parenting outcomes for children aged 0–4: A, hostile-ineffective parenting score; B, consistent parenting score; C, aversive parenting score. Data are drawn from the National Longitudinal Study of Children and Youth, waves 1–5.

satisfaction of 11.6 percent of a standard deviation. A full set of robustness checks for these results is presented in online Appendix table A5.

VI. Conclusion

In this paper we provide, to our knowledge, the first comprehensive analysis of a universal subsidized child care program, following its impact from child care use through employment and finally to children's and parents' outcomes. We uncover strong evidence of a shift into new child care use, although approximately one-third of the newly reported use appears to come from women who previously worked and had informal arrangements. The labor supply impact is strongly significant. Finally, we report striking evidence that children's outcomes have worsened since the program was introduced. We also find suggestive evidence that families we study became more stressed with the introduction of the program. This is manifested in increased aggressiveness and anxiety for the children; more hostile, less consistent parenting for the adults; and worse adult mental health and relationship satisfaction.

As discussed earlier, these results are subject to a number of interpretations that highlight the importance of future work in this area. Most important, it is not clear whether the negative child outcomes are merely short-run problems or presage long-run effects. In addition, we raise the puzzle of why families would take advantage of a policy that leads to worse child outcomes, worse parenting, and worse parental outcomes.³⁰ It is possible that the other unmeasured benefits of higher family incomes offset these costs. Alternatively, it is possible that families will learn that they are not better off in this new regime and that ultimately use of subsidized child care may fall. Once again, following the long-run evolution of these policy effects will be central to a full welfare analysis of the program.

Despite these qualifications, our results are particularly germane for ongoing policy debates in the United States and Canada. Our estimates do not speak to the efficacy of targeted child care subsidies for groups such as single mothers, but these are typically not the focus of the current debate. Instead, it is universal expansion of early child care/education that draws the most heated argument. Most directly, our evidence advises caution in Canada for other provinces considering adopting the Quebec child care model. It is possible that our findings are

³⁰ Blau (1991) explores the difference between choices made by parents and those recommended by child development experts, finding that parents do not appear willing to pay for apparently productive quality improvements.

short-run rather than long-run effects, but more evidence is needed before the program is adopted elsewhere.

In the United States, the debate is over universal preschool for 4-year-olds and possibly 3-year-olds. The motivation for these expansions is the positive cognitive returns for a limited set of experimental preschool interventions (see Anderson [2007] for a review of these experiments and a reinterpretation of the results) and to the public provision of preschool for low-income populations under the Head Start program (Currie and Thomas 1995). The effects may be quite different, however, for the balance of the population. Our results bring the social, behavioral, health, and family consequences of early nonparental care to a debate that has been preoccupied with cognitive results. Whether these negative findings for Canada would extend to preschool is an open question, but they highlight the importance for future study to consider wider arrays of developmental indices for more heterogeneous populations of children.

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